

The Art of Guard



Setting Security Contexts Using Policy Modules

Let's look at how to set default security contexts for files using policy modules. We will also use policy modules to set rules that will automatically transition the default type assigned to files and folders at the time of creation.

In the previous article, we had developed an SELinux policy module called *test* (version 1.1.1) for type enforcement. To do that, we had created a file called *test.te* which contained our allow rules and the new type (*lffy_t*) that we created in the system. We had written rules that allowed us to change the type of the */var/www/html/index.html* file from the default *httpd_sys_content_t* to *lffy_t*.

```
[root@vbg-work ~]# ls -lZ /var/www/html/index.html
-rw-r--r-- root:root unconfined_u:object_r:lffy_t:s0 /var/www/html/index.html
```

The only problem with this approach is that a restoration of file contexts would cause this file's security context to be relabelled back to *httpd_sys_content_t*. To check this, we can run the *restorecon* command:

```
[root@vbg-work ~]# restorecon -v /var/www/html/index.html
restorecon: reset /var/www/html/index.html context
unconfined_u:object_r:lffy_t:s0->system_u:object_r:httpd_sys_content_t:s0
```

```
[root@vbg-work ~]# ls -lZ /var/www/html/index.html
-rw-r--r-- root:root system_u:object_r:httpd_sys_content_t:s0 /var/www/html/index.html
[root@vbg-work ~]#
```

To avoid this issue and to set the default type of this file to *lffy_t*, we will need to add information to our policy module *test*. Entries that map files to desired security contexts are not specified in the *te*

(type enforcement) files.

A closer look at our development folder reveals that the *make* command (used to compile our source to a binary policy module) has already created two new files: *test.fc* and *test.if*. We will understand the purpose and the utility of the *fc* (file contexts) file in this article. A sample *fc* file, *example.fc*, is provided by the SELinux development RPM. The default location of this file is */usr/share/SELinux/devel/example.fc*. Let us have a look at the syntax:

```
[vbg@vbg-work test-selinux]$ cat example.fc
# myapp executable will have:
# # label: system_u:object_r:myapp_exec_t
# # MLS sensitivity: s0
# # MCS categories: <none>
#
/usr/sbin/myapp -- gen_context(system_u:object_r:myapp_exec_t,s0)
[root@vbg-work ~]#
```

Simply put, this file has three columns:

1. The first column contains the name of the file (or a regular expression pattern) to be labelled (since everything is a file in Linux).
2. The second column indicates the type of the file ('--' means any type of file).
3. The third column contains the default security context to be allotted to this file. It is very similar to file */etc/selinux/targeted/contexts/files/file_contexts* that we covered in the second article of this series. You may wish to revisit that article for the detailed explanation.